

INT362: CLOUD ARCHITECTURE AND IMPLEMENTATION-I

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: define the fundamental concepts of cloud computing and apache cloudstack, including cloud deployment models , cloudstack features and its architecture

CO2 :: describe cloudstack network architecture and deployment model of cloudstack

CO3 :: use the management server console to navigate and manage various tabs (dashboard, instance, storage, network.infrastructure etc.) within Cloudstack.

CO4 :: classify the configuration of basic and advanced zones, and create templates to set up the cloud infrastructure

CO5 :: interpret and manage comprehensive cloudstack networking to establish a private cloud

CO6 :: assess system requirements for Cloudstack storage, and configure primary and secondary storage systems

Unit I

Introduction to Cloud Computing : cloud computing concept, cloud computing models, deployment architecture

Apache Cloudstack : Introduction to Apache Cloudstack, features of cloudstack, Cloudstack architecture

Unit II

Cloudstack Network Architecture : Network Architecture overview, deployment model of cloudstack-basic mode, advanced mode, security groups, system VMs, virtual router

Unit III

Cloudstack Configuration : Management server console, dashboard tab, instance tab, storage tab, network tab, template tab, event tab, account tab, domain, infrastructure tab, project tab

Unit IV

Infrastructure set up : basic zone configuration, advanced zone configuration, creating template

Unit V

Cloudstack networking : zone, pods, cluster, hosts, Physical networking, Basic and advance zone, virtual network, creating a private cloud

Unit VI

Cloudstack Storage : System requirement and configuration, adding primary storage, secondary storage, adding secondary storage, Infra-as-Code based EC2 provisioning using Launch Templates and Auto Scaling groups

List of Practicals / Experiments:

List of practicals

- Installation of Apache CloudStack Management Server
- Apache CloudStack Management Server Setup
- CloudStack NFS server Setup
- CloudStack KVM Setup
- Creating Systemvm template
- Creating virtual private cloud
- Creating Basic zones
- Managing template and ISO

Text Books:

1. CLOUD COMPUTING: A PRACTICAL APPROACH FOR LEARNING AND IMPLEMENTATION by A. SRINIVASAN, J. SURESH, PEARSON

References:

1. APACHE CLOUDSTACK DOCUMENTATION by APACHE, N.A.

